College Event Feedback Analysis

---

1 Introduction

- 🔷 Task: Analyze student event feedback to uncover satisfaction trends and suggest improvements using survey data.
- 🔷 Skills Gained: Data cleaning, sentiment analysis, survey insights, charting, NLP.
- 🔷 Tools: Google Colab, pandas, TextBlob/VADER, seaborn, Google Forms (CSV).

2. Importing Libraries

In [ ]:
```
# import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Loading the datasets

In [ ]:
```
# load the dataset
df = pd.read_csv("/content/Event_feedback.csv")
```

4. Performing Exploratory Data Analysis (EDA)

In [ ]:
```
# we will start by seeing the first 5 rows
df.head()
```

Out[ ]:

	Student_ID	Event_Name	Organization_Rating	Content_Rating	Engagement_Level	Venu
0	1	Tech Fest	5	4	High	
1	2	Sports Meet	4	3	Medium	
2	3	Cultural Night	3	5	Low	
3	4	Workshop	5	4	High	
4	5	Guest Lecture	4	4	Medium	

In [ ]:

```
# then check out the last five rows
df.tail()
```

Out[ ]:

	Student_ID	Event_Name	Organization_Rating	Content_Rating	Engagement_Level	Ven
15	16	Tech Fest	4	4	Medium	
16	17	Sports Meet	5	5	High	
17	18	Cultural Night	3	3	Low	
18	19	Workshop	5	5	High	
19	20	Guest Lecture	4	4	Medium	

In [ ]:

```
df.columns
```

Out[ ]:

```
Index(['Student_ID', 'Event_Name', 'Organization_Rating', 'Content_Rating',
       'Engagement_Level', 'Venue_Rating', 'Speaker_Performance',
       'Event_Duration', 'Recommendation', 'Feedback'],
      dtype='object')
```

The columns names are good to go because they are uniform.

In [ ]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype 
---  ------                --------------  ----- 
 0   Student_ID            20 non-null     int64 
 1   Event_Name            20 non-null     object
 2   Organization_Rating   20 non-null     int64 
 3   Content_Rating        20 non-null     int64 
 4   Engagement_Level      20 non-null     object
 5   Venue_Rating          20 non-null     int64 
 6   Speaker_Performance   20 non-null     int64 
 7   Event_Duration        20 non-null     object
 8   Recommendation        20 non-null     object
 9   Feedback              20 non-null     object
dtypes: int64(5), object(5)
memory usage: 1.7+ KB
```

There are 5 integer columns and 5 character (object) columns.

In [ ]:
```
print(f"There are {df.shape[0]} rows and {df.shape[1]} columns")
```

```
There are 20 rows and 10 columns
```

In [ ]:
```
df.describe()
```

Out[ ]:

	Student_ID	Organization_Rating	Content_Rating	Venue_Rating	Speaker_Performan
count	20.00000	20.000000	20.00000	20.000000	20.0000
mean	10.50000	4.200000	4.15000	4.100000	4.3500
std	5.91608	0.767772	0.74516	0.967906	0.8127
min	1.00000	3.000000	3.00000	2.000000	3.0000
25%	5.75000	4.000000	4.00000	4.000000	4.0000
50%	10.50000	4.000000	4.00000	4.000000	5.0000
75%	15.25000	5.000000	5.00000	5.000000	5.0000
max	20.00000	5.000000	5.00000	5.000000	5.0000

The mean of Organization_Rating, Content_Rating, Venue_Rating, and Speaker_Performance is 4.20, 4.15, 4.10, and 4.35 respectively.

5. Data Cleaning

5.1 Missing values

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
Student_ID	0
Event_Name	0
Organization_Rating	0
Content_Rating	0
Engagement_Level	0
Venue_Rating	0
Speaker_Performance	0
Event_Duration	0
Recommendation	0
Feedback	0

dtype: int64

There is not a single missing value

5.2 Duplicates

In [ ]:
```
print(df.duplicated())
```

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
dtype: bool
```

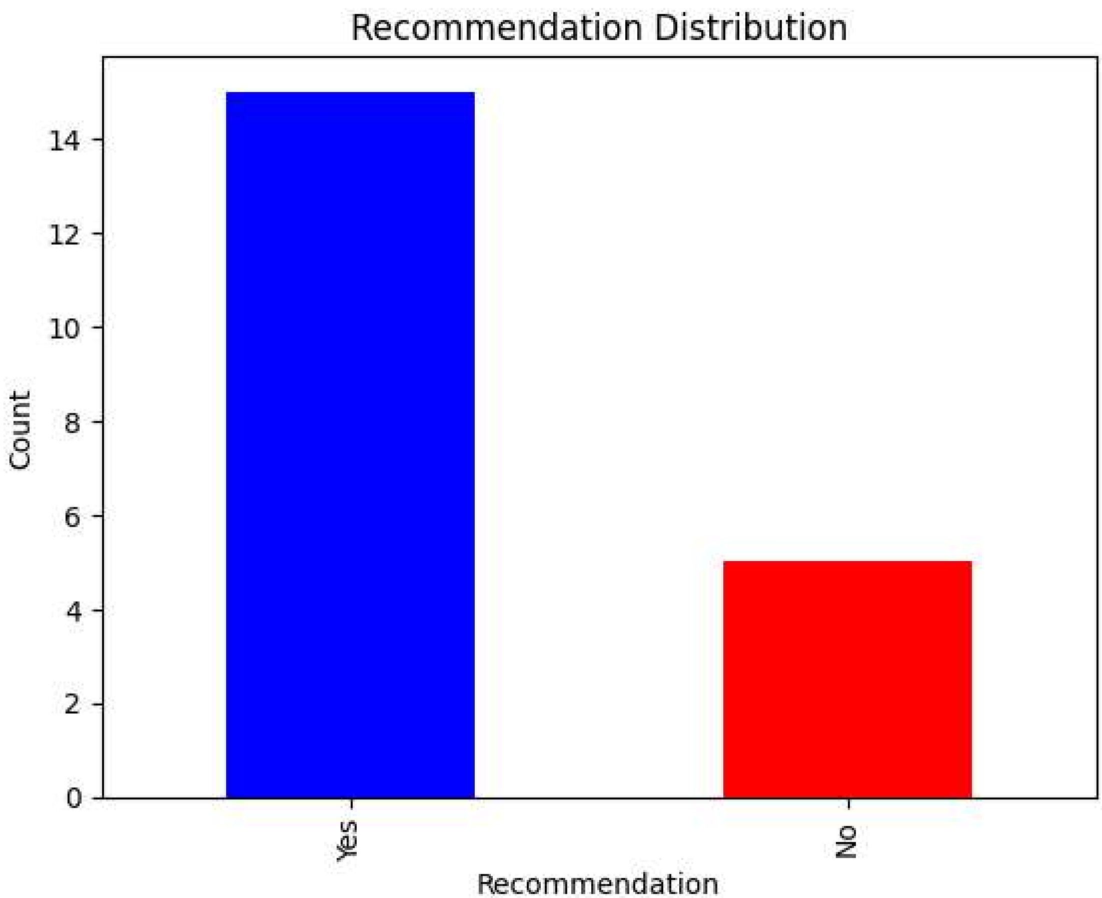
Each row is unique.

We are now going to data visualization section for insights

6. Data Visualization

6.1 Recommendation distribution

In [ ]:
```
# Recommendation distribution (Yes vs No)
recommendation_counts = df['Recommendation'].value_counts()
recommendation_counts.plot(kind='bar', color=['blue', 'red'])
plt.title('Recommendation Distribution')
plt.xlabel('Recommendation')
plt.ylabel('Count')
plt.show()
```

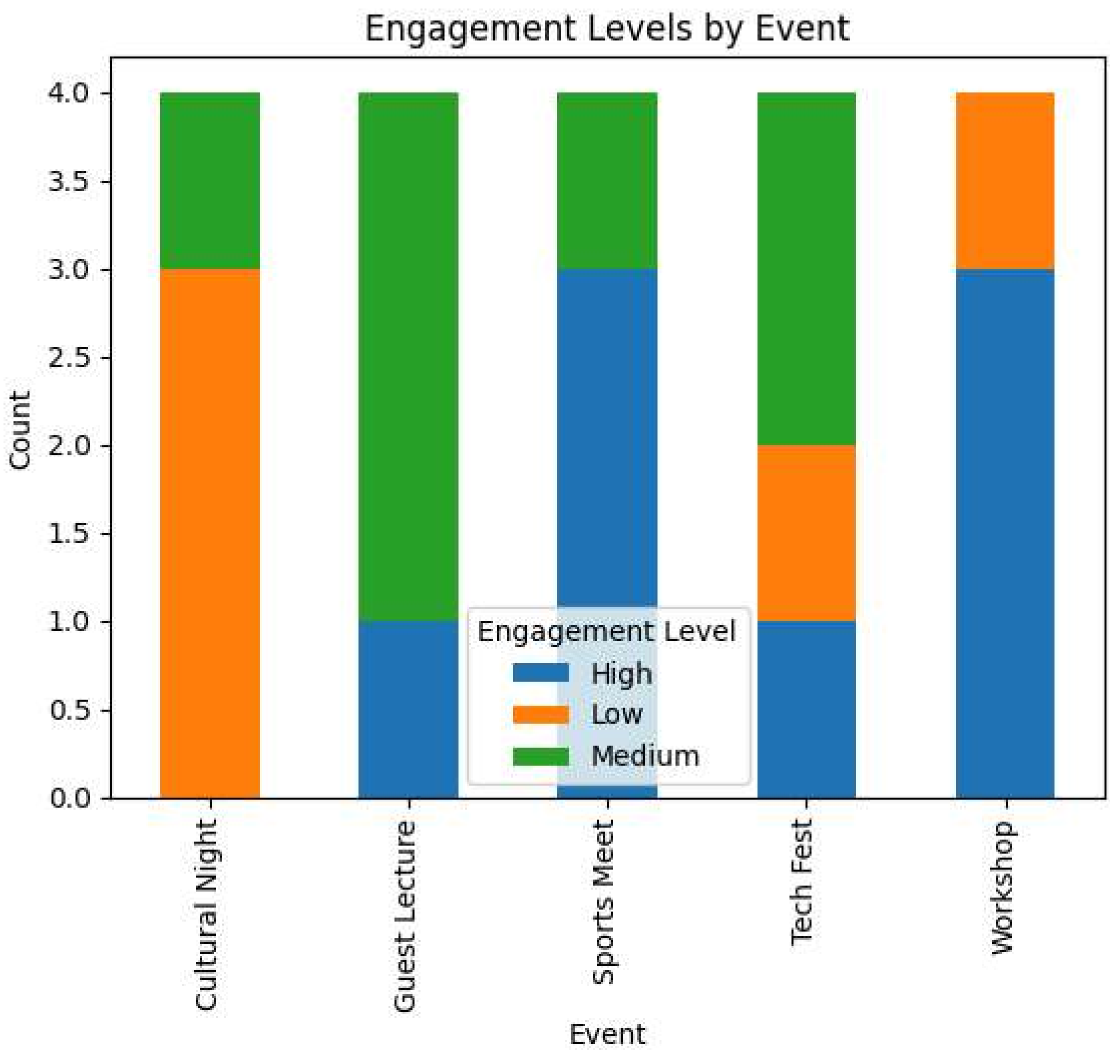


In [51]:
```
print("There are", recommendation_counts['No'], "'no' recommendations")
print("There are", recommendation_counts['Yes'], "'yes' recommendations")
# print(f"There are a total of {recommendation_counts} recommendation")
```

```
There are 5 'no' recommendations
There are 15 'yes' recommendations
```

6.2 Engagement distribution

In [ ]:
```python
# We will have to count the engagement levels for each event
engagement_by_event = df.groupby('Event_Name')['Engagement_Level'].value_counts().u
engagement_by_event.plot(kind='bar', stacked=True)
plt.title('Engagement Levels by Event')
plt.xlabel('Event')
plt.ylabel('Count')
plt.legend(title='Engagement Level')
plt.show()
```



In [52]:
```python
print(engagement_by_event)
```

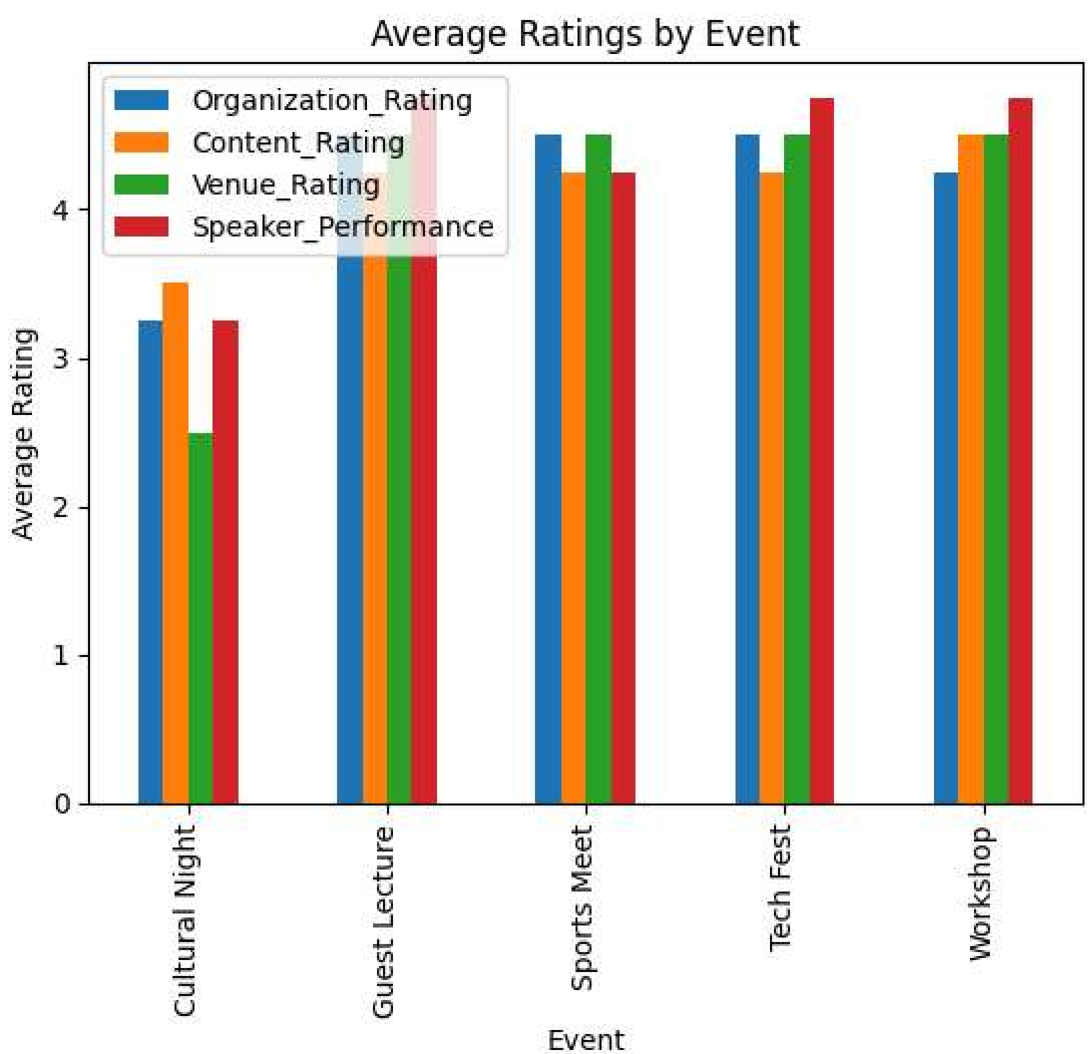
```
Engagement_Level  High  Low  Medium
Event_Name
Cultural Night     0.0  3.0     1.0
Guest Lecture      1.0  0.0     3.0
Sports Meet        3.0  0.0     1.0
Tech Fest          1.0  1.0     2.0
Workshop           3.0  1.0     0.0
```

6.3 Average ratings by Event

```
In [ ]: avg_ratings = df.groupby('Event_Name')[['Organization_Rating', 'Content_Rating', 'V
        print('Average Ratings by Event:\n', avg_ratings)
        avg_ratings.plot(kind='bar')
        plt.title('Average Ratings by Event')
        plt.xlabel('Event')
        plt.ylabel('Average Rating')
        plt.show()
```

```
Average Ratings by Event:
                  Organization_Rating  Content_Rating  Venue_Rating  \
Event_Name
Cultural Night                  3.25            3.50           2.5
Guest Lecture                   4.50            4.25           4.5
Sports Meet                     4.50            4.25           4.5
Tech Fest                       4.50            4.25           4.5
Workshop                        4.25            4.50           4.5

                Speaker_Performance
Event_Name
Cultural Night                  3.25
Guest Lecture                   4.75
Sports Meet                     4.25
Tech Fest                       4.75
Workshop                        4.75
```



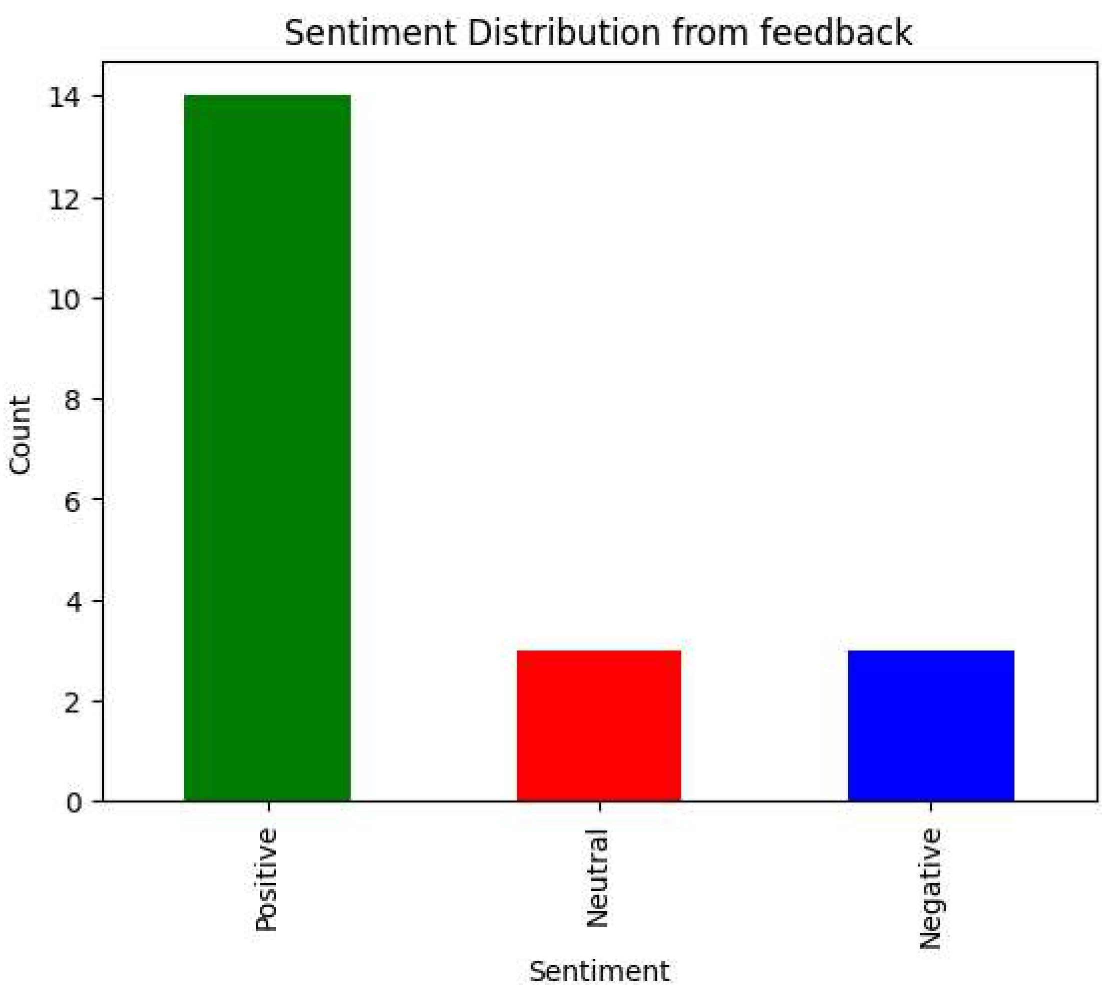
6.4 Sentiment Distribution from feedback

In [ ]:
```
from textblob import TextBlob

# function to get sentiment polarity
def get_sentiment(text):
  polarity = TextBlob(text).sentiment.polarity
  if polarity > 0.1:
    return 'Positive'
  elif polarity < -0.1:
    return 'Negative'
  else:
    return 'Neutral'

# Apply function to Feedback column
df['Sentiment'] = df['Feedback'].apply(get_sentiment)

# Sentiment distribution
sentiment_counts = df['Sentiment'].value_counts()
sentiment_counts.plot(kind='bar', color=['green', 'red', 'blue'])
plt.title('Sentiment Distribution from feedback')
plt.xlabel('Sentiment')
plt.ylabel('Count')
plt.show()
```



In [ ]:
```
print("There are:", sentiment_counts['Positive'], "positive feedback")
print("There are:", sentiment_counts['Negative'], "negative feedback")
print("There are:", sentiment_counts['Neutral'], "neutral feedback")
```

```
There are: 14 positive feedback
There are: 3 negative feedback
There are: 3 neutral feedback
```

6.5 Most common words in Student Feedback

In [ ]:
```
from wordcloud import WordCloud

# combine all feedback into one text
feedback_text = ' '.join(df['Feedback'].astype(str))

# create word cloud
wordcloud = WordCloud(width=800, height=400, background_color='Black').generate(fee

# show wordcloud
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Most common words in Student Feedback')
plt.show()
```

Most common words in Student Feedback

We can see that fun and well are the most common words.

6.6 Constructive Feedback

In [ ]:
```
from wordcloud import STOPWORDS

#filter only negative and neutral feedback
constructive_feedback = df[df['Sentiment'].isin(['Negative', 'Neutral'])]['Feedback
```

```
# Combine all improvement feedback into one string
text_improvement = " ".join(constructive_feedback.astype(str))

# Generate Word Cloud
wordcloud_improvement = WordCloud(width=800, height=400, background_color='black').

# Plot Word Cloud
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_improvement, interpolation='bilinear')
plt.axis('off')
plt.title('Constructive Feedback')
plt.show()
```

Constructive Feedback

Crowded and Engaging are the most common word in the constructive feedback section

7. Conclusion

The feedback shows that most students enjoyed the college events and were satisfied with how they were organized. Ratings for speakers, venues, and overall experience were generally positive, and many students said they would recommend the events to others though there was constructive feedback